

Lecture 3: Stability

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English version of slides by ©Bo Wahlberg





Outline

Introduction

Recap: PID control

Stability

Root Locus

The Nyquist stability criterion

Today's Ordlista

English	Svenska
open-loop control	öppen styrning
sensitivity function	känslighetsfunktion
open-loop system	öppet system, kretsförstärkning
root locus	rotort
asymptotes	asymptoter
starting points	startpunkter
ending points	ändpunkter
intersection point	skärningspunkten
branches	grenar
Nyquist stability criterion	Nyquistkriteriet
argument variation principle	argumentvariationsprincipen

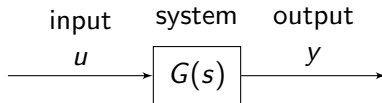
In the previous lecture..

- PID control

Today:

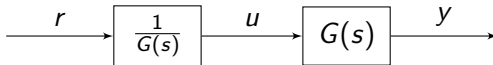
- Stability
 - Root locus
 - Nyquist stability criterion

Recap: What is reglerteknik?



The goal is $y = r$ (reference signal), despite disturbances and limitations.

Idea: *open-loop control*, inversion

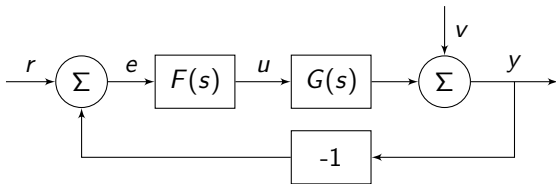


Problems:

- ▶ Does not take into account disturbances and model errors (+ initial values)
- ▶ Big input $U(s) = \frac{1}{G(s)}R(s)$ (if G has low gain)
- ▶ Stable?

Recap: What is reglerteknik?

Feedback from the control error $e = r - y$.



► $r \mapsto y$: $G_c = \frac{GF}{1+GF}$ = closed-loop transfer function

► $r \mapsto u$: $G_{ru} = \frac{F}{1+GF}$

► $v \mapsto y$: $S := G_{vy} = \frac{1}{1+GF}$ = sensitivity function

Obs: $G_o = GF$ = open-loop transfer function (at the denominator of both S and G_c)



- $G_c = \frac{KG}{1+KG} \approx 1$ if K big
- $G_{ru} = \frac{K}{1+KG} = \frac{KG}{1+KG} \frac{1}{G} \approx \frac{1}{G}$ if K big

“Inversion with the help of feedback!”

Problem with the stability! (LAB 1)

PID control: PI control

PI control:

PI control: $u(t) = K \left(e(t) + \frac{1}{T_I} \int_{t_0}^t e(\tau) d\tau \right) \Rightarrow F(s) = K \left(1 + \frac{1}{T_I} \frac{1}{s} \right)$

Observe that:

$$F(0) = \infty \implies G_c(0) = \frac{F(0)G(0)}{1 + F(0)G(0)} = 1$$

i.e. no stationary error, $\lim_{t \rightarrow \infty} e(t) = 0$

Why? See the final value theorem [(A.12), p. 246]

Problems:

- Problem with the stability
- Problem if the control signal, u , saturates (integrator windup)

PID control: PD control

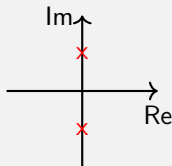
PD control: $u(t) = K \left(e(t) + T_D \frac{de(t)}{dt} \right) \Rightarrow F(s) = K(1 + T_D s)$

Example (Inverted pendulum)

The system has a transfer function $\approx$ in the form $G(s) = \frac{1}{s^2 - g/l}$

$$G_c(s) = \frac{G(s)F(s)}{1 + G(s)F(s)} = \frac{K(1 + T_D s)}{s^2 + K T_D s - g/l + K}$$

- **P control**, $T_D = 0$: **Does not work**
(the closed-loop system oscillates, poles on Im axis)
- **PD control**, $T_D \neq 0$: If we choose $T_D = 2/K$,
 $K = 1 + g/l \Rightarrow (s + 1)^2 = 0$, i.e., two poles in -1 : **OK!**
The D -term provides stability (dampens the oscillations)!





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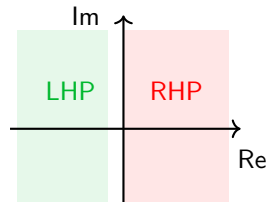
Stability

Root Locus

The Nyquist stability criterion

Stability: How does one determine stability?

Asymptotic stability = Poles strictly in the LHP
 $\Rightarrow$ input-output stable [Sats 2.1, 2.2]



1. Calculate the poles of the closed-loop system

Example (2nd degree polynomial)

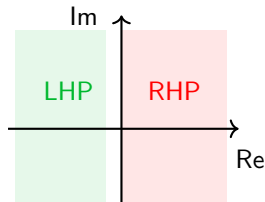
$$(s - a)(s - b) = s^2 - (a + b)s + ab,$$

Stable poles $\Leftrightarrow a$ och b in LHP $\Leftrightarrow -(a + b) > 0$ and $ab > 0$, i.e. the polynomial has positive coefficients. **Example:** $a = -1$, $b = -2 \Rightarrow (s + 1)(s + 2) = s^2 + 3s + 2$

- 2.
- Routh's algorithm [Appendix 2.A, p. 45]
 - Root Locus [Chapter 3.7, Appendix 3.A] [+Chapter 3.6 AH]
 - Nyquist stability criterion [Chapter 3.8] [+Chapter 11.1 AH]

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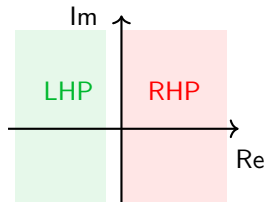
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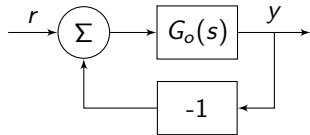
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 - **Root Locus** [Chapter 3.7, Appendix 3.A] [+Chapter 3.6 AH]
 - **Nyquist stability criterion** [Chapter 3.8] [+Chapter 11.1 AH]

Root Locus

$G_o(s)$ is the open-loop system, $G_c(s)$ is the closed-loop system

$$Y(s) = \frac{G_o(s)}{\underbrace{1 + G_o(s)}_{G_c(s)}} R(s)$$



Assume that

$$G_o(s) = K \frac{Q(s)}{P(s)} = K \frac{s^m + b_1 s^{m-1} + \dots + b_m}{s^n + a_1 s^{n-1} + \dots + a_n} \Rightarrow G_c(s) = \frac{K \frac{Q(s)}{P(s)}}{1 + K \frac{Q(s)}{P(s)}} = \frac{KQ(s)}{P(s) + KQ(s)}$$

The poles of the **closed-loop system** are given by

$$P(s) + KQ(s) = 0 \quad (*)$$

Root locus: A plot of the poles, i.e., the roots of (*), as a function of K , $0 < K < \infty$.

Idea: I want to predict where the poles are going as a function of K

Root Locus: Step-by-step

0. (#roots/branches). Since $(*)$ has degree n , there are n roots for each value of K
 $\Rightarrow$ we say that the root locus has **n branches**

Note: roots are either real or complex conjugate $\Rightarrow$ **root locus is symmetric w.r.t. Re-axis**

I. (starting points). When $K = 0$, $(*)$ becomes $P(s) = 0 \Rightarrow$ The n branches of the root locus start at the roots of $P(s) \Rightarrow$ **starting points = zeros of $P(s)$**

II. (ending points). Of these n branches, m will end (for $K = \infty$) at the roots of $Q(s) \Rightarrow$ **ending points = zeros of $Q(s)$**

III. (asymptotes). If $m < n$, the remaining $n - m$ branches will go towards $\infty \Rightarrow$ the curves the root locus approaches as $s \rightarrow \infty$ are called **asymptotes** [Resultat 3.1]

- For $s \rightarrow \infty$, a good approximation of $(*)$ is: $(s + \frac{a_1 - b_1}{n - m})^{n - m} = -K$
- **direction** $\Rightarrow \arg(s + \frac{a_1 - b_1}{n - m}) = \frac{\pi}{n - m} + 2k \frac{\pi}{n - m}, k = 0, 1, 2, \dots$
- **intersection points** $\Rightarrow -\frac{a_1 - b_1}{n - m}$

IV. (roots on Re-axis?). Root locus is found on a given section of the Re-axis only if the **total # of poles and zeros of G_o to the right of the section is odd** [Resultat 3.2]

V. (roots on Im-axis?). Assume $s = i\omega$ and **solve $(*)$ w.r.t. ω and K** . [Resultat 3.2]

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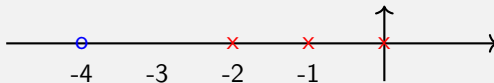
Root Locus: Example

Example

Assume $G_o(s) = \frac{K(s+4)}{s(s+1)(s+2)}$. This gives the equation (*) in the form

$$\underbrace{s(s+1)(s+2)}_{P(s)} + K \underbrace{(s+4)}_{Q(s)} = 0 \iff \frac{s(s+1)(s+2)}{(s+4)} = -K$$

0. (*) has degree $n = 3 \Rightarrow$ **3 branches**
- I. Starting points, $K = 0$: roots of P . In our case, we obtain **$s = 0, -1, -2$** . We mark these with crosses in the plot.
- II. Ending points, $K = \infty$: roots of Q . In our case, **$s = -4$** . We mark these with circles in the plot.



Root Locus: Example

Example (cont'd.)

- III. There are $n - m = 2$ asymptotes.

How do we determine angle/intersections?

- *Method 1. Study*

$$\arg\left(\frac{s(s+1)(s+2)}{(s+4)}\right) = \arg(-K) = \pi + 2\pi k, \quad k = 0, 1, \dots$$

$$\Rightarrow \arg s + \arg(s+1) + \arg(s+2) - \arg(s+4) = \pi + 2\pi n$$

If s is large, all angles (arguments) will be approximately the same size:

$$\Rightarrow 3\varphi - \varphi = \pi + 2\pi k \Rightarrow \varphi = \frac{\pi}{2} + \pi k$$

In our case, we have one branch that goes towards $\pi/2$ and one that goes towards $\frac{3}{2}\pi$

Root Locus: Example

Example (cont'd.)

III. There are $n - m = 2$ asymptotes.

How do we determine angle/intersections?

- *Method 2.* A more detailed analysis (uses binomial theorem/or binomial expansion):

$$\begin{aligned} \frac{s(s+1)(s+2)}{(s+4)} &= s^2 + (1+2-4)s + \frac{6s}{s+4} \\ &= \left(s + \frac{(1+2-4)}{2}\right)^2 - \frac{1}{4} + \frac{6s}{s+4} = \left(s - \frac{1}{2}\right)^2 - \underbrace{\frac{1}{4} + \frac{6s}{s+4}}_{\rightarrow -1/4+6} \end{aligned}$$

For large s the 1st term dominates, which for large K gives the equation

$$\left(s - \frac{1}{2}\right)^2 = -K$$

Describes the asymptotes in more detail! The roots are placed in $s = 1/2 \pm \sqrt{K}i$

Root Locus: Example

Example (cont'd.)

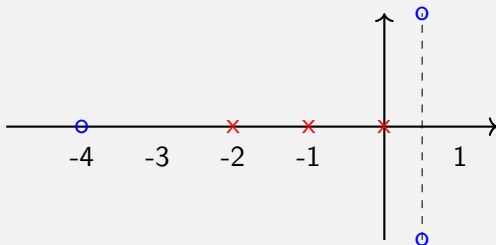
III. Conclusion: There are 2 asymptotes

- that start at the *intersection point*:

$$-\frac{a_1 - b_1}{n - m} = -\frac{1 + 2 - (-4)}{2} = \frac{1}{2}$$

- ...and have *directions*:

$$\frac{\pi}{n - m} + 2k\frac{\pi}{n - m} \quad (k = 0, 1, 2, \dots) = \frac{\pi}{2}, \frac{3\pi}{2}$$



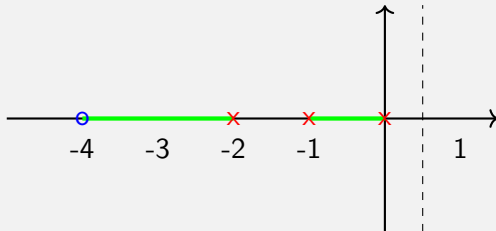
Root Locus: Example

Example (cont'd.)

IV. Where on the Re-axis can there be roots?

- Example: Can we have a pole in -5 ? No (see rule)
- Example: Can we have a pole in $s = -3$? Yes.

$$\text{Solve } -K = \frac{-3(-2)(-1)}{-3+4} = -6 \Rightarrow K = 6$$



Root Locus: Example

Example (cont'd.)

V. Where on the Im-axis can there be roots?

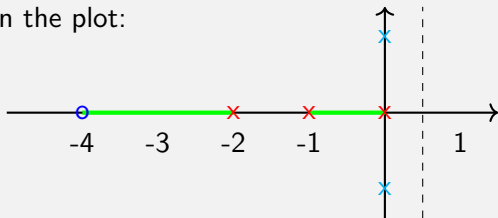
Assume that $s = i\omega$ in (*) and solve with respect to ω and K .

$$s^3 + 3s^2 + 2s + Ks + 4K = 0 \Rightarrow -i\omega^3 - 3\omega^2 + 2i\omega + Ki\omega + 4K = 0$$

- The real part: $-3\omega^2 + 4K = 0 \Rightarrow \omega^2 = \frac{4}{3}K$
- The imaginary part: $-\omega^3 + 2\omega + K\omega = 0 \Rightarrow \begin{cases} \omega = 0 \\ \omega^2 = 2 + K \end{cases}$
- The first solution is $\omega = 0, K = 0$.

The second, $\omega^2 = 2 + K$, gives $2 + K = \frac{4}{3}K$, i.e., $\begin{cases} K = 6 \\ \omega = \pm\sqrt{8} \approx \pm 2.8 \end{cases}$

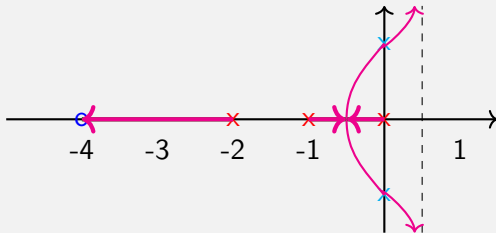
We mark this point on the plot:



Root Locus: Example

Example (cont'd.)

Conclusion



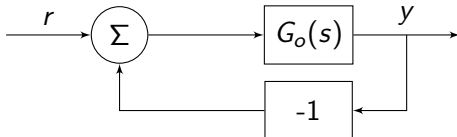
Stable for $K < 6$. Oscillatory when K approaches 6. For large K , the poles are given by the zero points of the open-loop system and relative degrees (zero points at infinity).

Final observations:

K is often the gain in a P controller. However, the method is more general.

An important tool for getting an idea of what happens when you change a parameter.

The Nyquist stability criterion



$$\Rightarrow G_c = \frac{G_o}{1 + G_o}$$

Poles of $G_c(s)$ are given by the zeros of $1 + G_o(s)$!*

Theorem (Nyquist stability criterion)

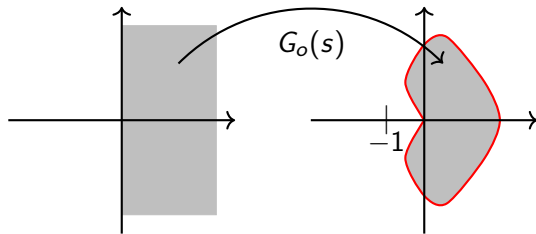
*# of unstable poles of $G_o(s)$ = # of unstable poles of $G_c(s)$
+ # of anticlockwise encirclements of -1 made by the Nyquist plot of $G_o(s)$*

* Is this easy to do/look at? **Homework:** Try out pzmap of $G_o = \text{tf}([1 \ 3 \ 5 \ 7], [2 \ 4 \ 6 \ 8 \ 2])$ and of $G_o + 1$, and then look at the corresponding Nyquist plots!

The Nyquist stability criterion

Is there any $s \in \text{RHP}$ such that $G_o(s) = -1$?
If so, the closed-loop system $G_c(s)$ is unstable!

Idea: Try all the $s \in \text{RHP}$



In the above figure, there is no $s \in \text{RHP}$ such that $G_o(s) = -1$.
 $\Rightarrow$ Stable closed-loop system!

The Nyquist stability criterion

It is smarter to just check the margin, i.e., the Nyquist contour..

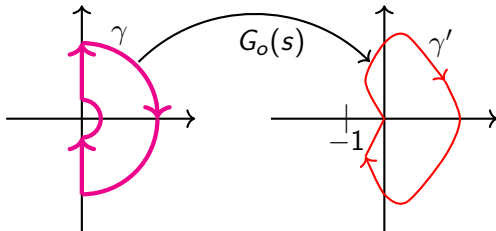
Paths for γ :

$$s = i\omega, \omega \in (0, \infty)$$

$$s = Re^{i\theta}, R \rightarrow \infty, \theta : \frac{\pi}{2} \rightarrow -\frac{\pi}{2}$$

$$s = i\omega, \omega \in (-\infty, 0)$$

$$s = re^{i\theta}, r \rightarrow 0, \theta : -\frac{\pi}{2} \rightarrow \frac{\pi}{2}$$



Use **Cauchy's argument variation principle***:

$$\begin{aligned}
 & -\# \text{ anticlockwise encirclements of } -1 \text{ by } G_o(s) \text{ **} \\
 & = \underbrace{\# \text{ zeros of } 1 + G_o(s) \text{ in RHP}}_{\# \text{ unstable poles of } G_c(s)} - \underbrace{\# \text{ poles of } 1 + G_o(s) \text{ in RHP}}_{\# \text{ unstable poles of } G_o(s)}
 \end{aligned}$$

* How many zeros does a rational function $G_o(\cdot)$ have in a region?

** The argument principle (i.e., $\# \text{ anticlockwise encirclements} = \# \text{ zeros} - \# \text{ poles}$) is for a contour that goes anticlockwise! To obtain the direction that we drew for the contour, we have to add a minus sign.



Assume that $G_o(s)$ is stable (poles in $s = 0$ are OK).
Then $G_c(s)$ is asymptotically stable if γ' does not encircle -1 .

- Nyquist criterion: We used the **argument principle**
- The most important part of the γ' -curve is $G_o(s)$, $s = i\omega$, $r < \omega < R$.
- $G_o(\infty) = 0$ and $G_o(0) \approx K/s^p$ where p is the number of poles in the origin.